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You can now make this inexpensive and easy-to-make LED bike light (primarily to support the stock headlight) that also has a strobe light

Circuit and working

If you are wondering why use a slightly expensive Digispark instead of cheaper discrete components, the reason is the light's two modes—steady glow and strobe. That means when knob of the

Likewise, Digispark's P1 I/O pin goes to the gate of the lamp driver MOSFET IRL520N (IRF1). The 100-ohm resistor R1 is the gate resistor of IRF1 whereas the 10k resistor R2 works as a pull-down resistor. These two resistors are essential because the gate of the MOSFET is basically a capacitor. So, when a voltage is applied to the gate without the gate resistor, the resultant inrush current can damage the microcontroller's I/O if it is unable to quickly source that much current. Here, R1 limits the maximum current and R2 assures that IRF1 is always in a known state. Fig. 3 shows a casual oscilloscope snap captured while poked into TP1.

If you are using a non-USB version

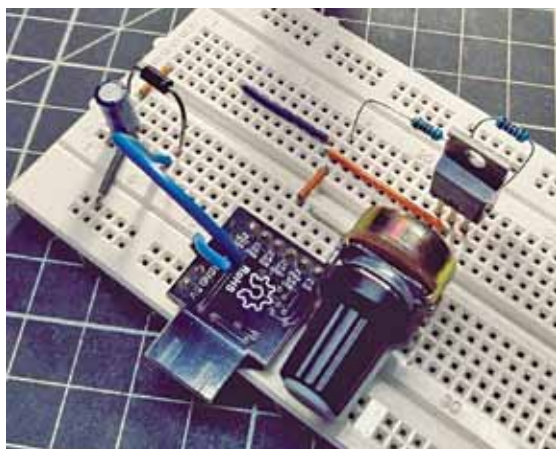


Fig. 1: Author's prototype

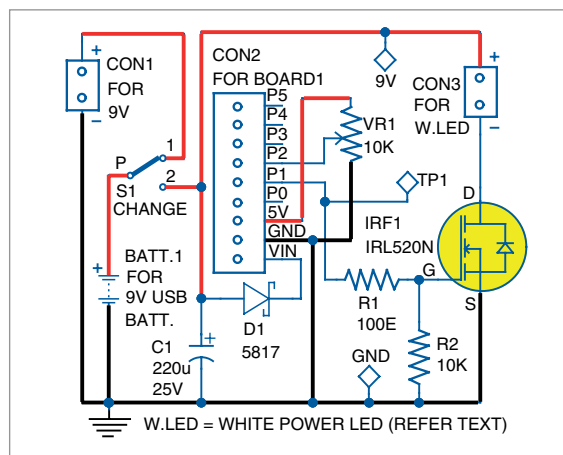


Fig. 2: Circuit diagram of LED bike light

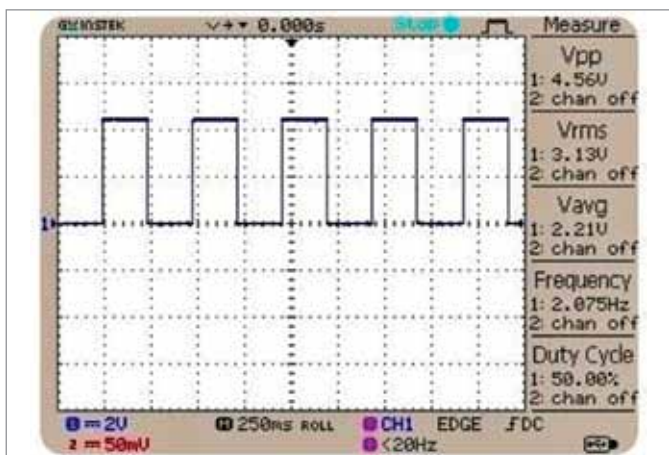


Fig. 3: Oscilloscope snap (see text)